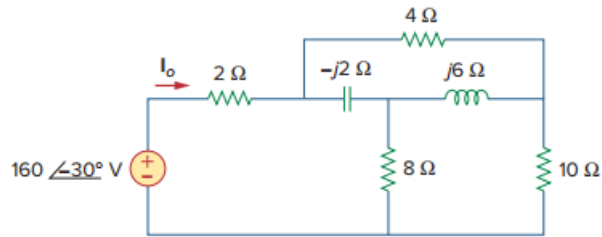


Problem

Find I_o in the circuit of Fig. 9.60.

Figure 9.60:



Step-by-step solution

Step 1 of 5

Refer to the Figure 9.60 in the textbook.

Converting the Δ network consisting $-j2\Omega$, $j6\Omega$ and 4Ω to Y network,

$$Z_1 = -j2\Omega$$

$$Z_2 = j6\Omega$$

$$Z_3 = 4\Omega$$

Calculate the impedance Z_a .

$$\begin{aligned} Z_a &= \frac{Z_1 Z_2}{Z_1 + Z_2 + Z_3} \\ &= \frac{(-j2)(j6)}{-j2 + j6 + 4} \\ &= \frac{12}{4 + j4} \end{aligned}$$

Simplify further.

$$\begin{aligned} Z_a &= \frac{12}{4 + j4} \times \frac{4 - j4}{4 - j4} \\ &= \frac{48 - j48}{32} \\ Z_a &= (1.5 - j1.5)\Omega \end{aligned}$$

Step 2 of 5

Calculate the impedance Z_b .

$$\begin{aligned} Z_b &= \frac{Z_2 Z_3}{Z_1 + Z_2 + Z_3} \\ &= \frac{(j6)(4)}{-j2 + j6 + 4} \\ Z_b &= \frac{j24}{4 + j4} \end{aligned}$$

Simplify further.

$$\begin{aligned} Z_b &= \frac{j24}{4 + j4} \times \frac{4 - j4}{4 - j4} \\ &= \frac{96 + j96}{32} \\ Z_b &= (3 + j3)\Omega \end{aligned}$$

Step 3 of 5

Calculate the impedance Z_c .

$$\begin{aligned} Z_c &= \frac{Z_3 Z_1}{Z_1 + Z_2 + Z_3} \\ &= \frac{(4)(-j2)}{-j2 + j6 + 4} \\ &= \frac{-j8}{4 + j4} \end{aligned}$$

Simplify further.

$$\begin{aligned} Z_c &= \frac{-j8}{4 + j4} \times \frac{4 - j4}{4 - j4} \\ &= \frac{-32 - j32}{32} \\ &= (-1 - j1)\Omega \\ Z_c &= \frac{-32 - j32}{32} \\ Z_c &= (-1 - j1)\Omega \end{aligned}$$

Therefore, the reduced circuit diagram after the delta-wye conversion is shown in Figure 1.

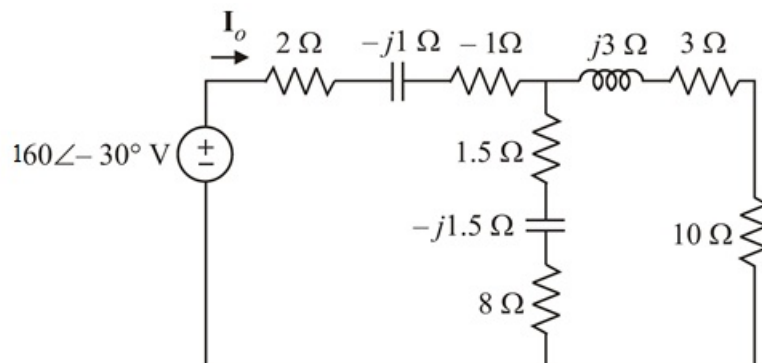


Figure 1

Step 4 of 5

Calculate the total impedance Z of the circuit diagram in the Figure 2.

$$\begin{aligned}Z_{eq} &= (2 - j1 - 1) + [(1.5 - j1.5 + 8) \parallel (j3 + 3 + 10)] \\&= (1 - j1) + [(9.5 - j1.5) \parallel (13 + j3)] \\&= (1 - j1) + \frac{(9.5 - j1.5)(13 + j3)}{(9.5 - j1.5) + (13 + j3)} \\&= (1 - j1) + \frac{128 + j9}{22.5 + j1.5} \\Z_{eq} &= 1 - j1 + 5.69 + j0.0206 \\&= (6.69 - j0.9794) \\&= 6.7613 \angle -8.33^\circ \Omega\end{aligned}$$

Step 5 of 5

Calculate the current I_0 supplied by the voltage source.

$$\begin{aligned}I_0 &= \frac{V}{Z} \\&= \frac{160 \angle -30^\circ}{6.7613 \angle -8.33^\circ \Omega} \\&= 23.66 \angle -21.67^\circ \text{ A}\end{aligned}$$

Therefore, the current I_0 supplied by the voltage source is $23.66 \angle -21.67^\circ \text{ A}$.